

MTU_ValueService **Technical Documentation**

System Description

ADEC and SAM connection interface

Series 2000 unit-pump system, Series 4000

Application: Genset (SAM “advanced”)

System documentation

E532304/00E



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The Commissioning Note information serves as a basis for the contractually agreed logistic support (warranty, spare parts, etc.).



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Technical Information Management
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88040 Friedrichshafen
GERMANY

Commissioning Note

Please use block capitals!



Engine No.:	MTU works order No.:
Engine model:	Date put into operation:
Installation site:	Vessel/type/class / Shipyard:
End user`s address:	
Remarks:	

**Commission-
ing Note**

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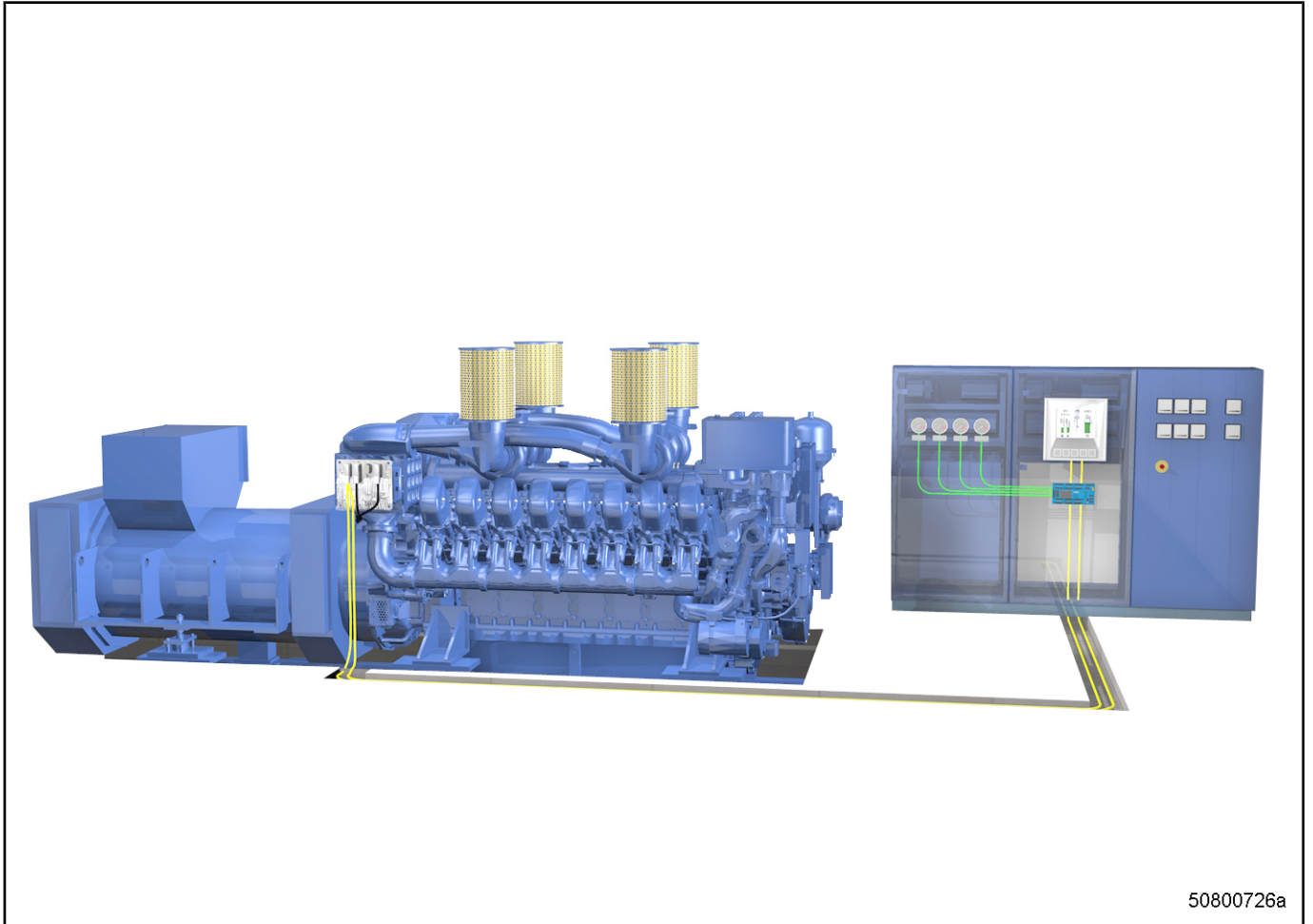
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1 Functional Description

1.1 Purpose, Options and Features

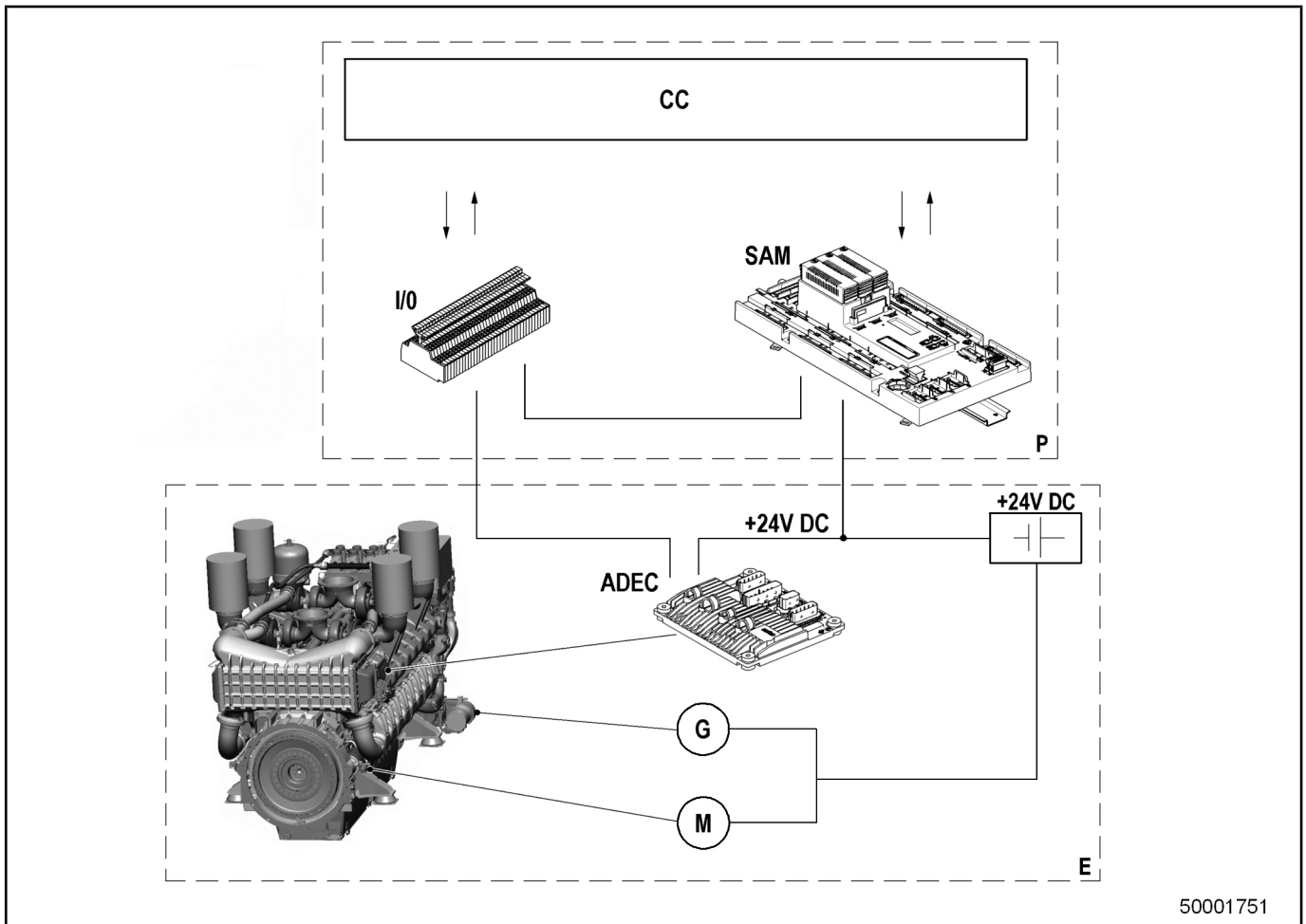
1.1.1 Use

Functions



- Control of the engine.
- Monitoring of operating states.
- Closed-loop control of fuel injection and engine speed (depending on operating state).
- Indication of faulty operating states (display SAM).

Product summary



CC Customer Controller — System
 I/O Terminal block (OEM) for inputs and outputs
 SAM Service and Automation Module (SAM)

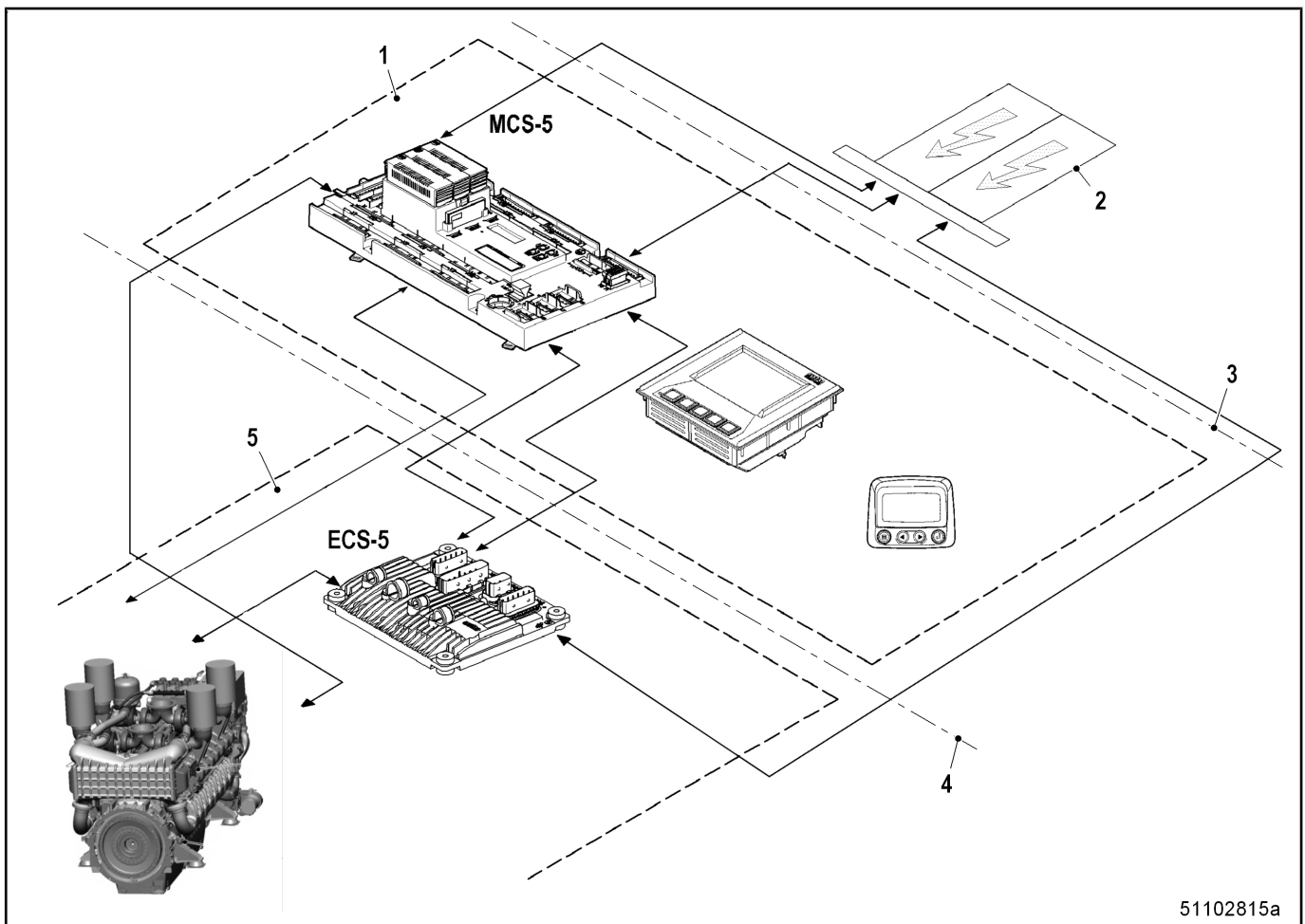
P Plant
 E Engine
 ADEC Engine Control Unit

G Battery-charging generator
 M Starter

Features

- Electronic engine management and control.
- Monitoring of inadmissible engine operating states.
- Display of fault messages and fault codes.
- Connecting cable for power supply to Engine Control Unit.
- Connecting cable for connection to a higher-level genset control system.
- Hardware interfaces to a higher-level control system (option).
- Inputs for customer sensors (option).
- Engine safety features including engine shutdown.
- Integral fault diagnostic system ITS.
- Integral load profile recorder.
- Speed droop switching possible with engine running.
- Straightforward Engine Control Unit replacement.
 - Engine data and interface data are saved in the SAM.
 - Complete automatic software download following connection of a new, unprogrammed Engine Control Unit.
- ECMS — Engine Side Condition Management System
 - Automatic engine power reduction as a function of:
 - Intake air temperature
 - Operating site (e.g. altitude above sea level)
 - Other parameters

Design of the overall system



51102815a

- 1 Monitoring and Control System MCS-5
(DIS 10/PowerView option)
2 CAN bus between ECS-5 and MCS-5

- 3 Engine Control System ECS-5
4 Interface for external control

- 5 External control

The design of the overall system comprising the ECS-5 and MCS-5 subsystems depends on customer requirements and the higher-level control system.

Design of Engine Control System ECS-5

The Engine Control System comprises the following devices:

- Engine Control Unit
- Sensors on the engine
- Actuators on the engine
- Injectors on the engine
- Wiring harnesses on the engine